Name: _____ Date: _____

PS-03: $\varepsilon - \delta$ Proofs, Part III

Topics
• $\varepsilon - \delta$ Limits at Infinity

Directions: You can collaborate with classmates so long as you attribute any notable **lightbulb moments**. This will be graded for correctness.

1. $\lim\limits_{t\to\infty} \dfrac{t^2}{t-t^2} = -1$
2. $\lim\limits_{x\to\infty} \dfrac{3x^2}{x^2-8x+15} = 3$
3. $\lim\limits_{x\to\infty} \dfrac{3\sqrt{x^3}+3x}{\sqrt{2x^3}} = \dfrac{3}{\sqrt{2}}$
4. $\lim\limits_{t\to\infty} \dfrac{\sqrt{2t+1}}{t+4} = 0$

Solution 1

Solution 1.1

Let $\varepsilon > 0$ and choose $M = 1 + \dfrac{1}{\varepsilon} > 1$. Then, if $t > M$, we have

$$
\begin{aligned}
|f(t) - L| &= \left| \frac{t^2}{t-t^2} - (-1) \right| \\
&= \left| \frac{t^2}{t-t^2} + 1 \right| \\
&= \left| \frac{t^2 + (t-t^2)}{t-t^2} \right| \\
&= \left| \frac{t}{t(1-t)} \right| && \text{since } t > M \geq 0 \\
&= \frac{1}{|1-t|} \\
&= \frac{1}{|t-1|} \\
&= \frac{1}{t-1} && \text{since } t > M > 1 \\
&< \frac{1}{M-1} && \text{since } \frac{1}{x-1} \text{ is decreasing for } x > 1 \\
&= \frac{1}{(1+\frac{1}{\varepsilon})-1} \\
&= \varepsilon \qquad \square
\end{aligned}
$$

Solution 1.2

Let $\varepsilon > 0$ and choose $M = \ldots$. Then, if $t > M$, we have

$$\begin{aligned}
|f(t) - L| &= \left| \frac{t^2}{t - t^2} - (-1) \right| \\
&= \left| \frac{t^2}{t - t^2} + 1 \right| \\
&= \left| \frac{t^2 + (t - t^2)}{t - t^2} \right| \\
&= \left| \frac{t}{t(1-t)} \right| && \text{since } t > M \geq 0 \\
&= \frac{1}{|1 - t|} \\
&= \frac{1}{|t - 1|} \\
&= \frac{1}{t - 1} && \text{since } t > M > 1 \\
&< \frac{1}{M - 1} && \text{since } \frac{1}{x-1} \text{ is decreasing for } x > 1 \\
&= \frac{1}{(1 + \frac{1}{\varepsilon}) - 1} \\
&= \varepsilon \qquad \square
\end{aligned}$$

Solution 2

Solution 2.1

Let $\varepsilon > 0$ and choose $M = \max\left(16, \dfrac{48}{\varepsilon}\right)$. Then, if $x > M$, we have

$$\begin{aligned}
|f(x) - L| &= \left| \frac{3x^2}{x^2 - 8x + 15} - 3 \right| \\
&= \left| \frac{3x^2 - 3(x^2 - 8x + 15)}{x^2 - 8x + 15} \right| \\
&= \left| \frac{24x - 45}{x^2 - 8x + 15} \right| \\
&= \frac{|24x - 45|}{|x^2 - 8x + 15|}
\end{aligned}$$

Considering the numerator and denominator separately, we have

$$|24x - 45| > 0 \quad \text{if } x > 2$$

and

$$\left|x^2 - 8x + 15\right| = |(x - 3)(x - 5)| > 0 \quad \text{if } x > 5$$

Let's assume, then, that $M \geq 5$. Therefore,

$$\begin{aligned} |f(x) - L| = \ldots &= \frac{|24x - 45|}{|x^2 - 8x + 15|} \\ &= \frac{24x - 45}{x^2 - 8x + 15} \\ &< \frac{24x}{x^2 - 8x + 15} \end{aligned}$$

Let's bound the denominator:

$$x^2 - 8x + 15 > \frac{1}{2}x^2 \iff \frac{1}{2}x^2 - 8x + 15 > 0$$

To find the zeros of the above quadratic expression, use the quadratic formula:

$$x = \frac{16 \pm \sqrt{156 - 4(30)}}{2} = \frac{16 \pm \sqrt{244}}{2} = \frac{16 \pm 2\sqrt{61}}{2} = 8 \pm \sqrt{61}$$

Now, the larger root is $8 + \sqrt{61} < 16$. Therefore, for $x > 16$, we have

$$x^2 - 8x + 15 > \frac{1}{2}x^2$$

Therefore, we'll assume $M \geq 16$ and can say

$$\begin{aligned} |f(x) - L| = \ldots &= \frac{24x - 45}{x^2 - 8x + 15} \\ &< \frac{24x}{\frac{1}{2}x^2} \\ &= \frac{48}{x} \\ &= \frac{48}{48/\varepsilon} \\ &= \varepsilon \qquad \square \end{aligned}$$

Solution 2.2

Setup:

$$\left| \frac{3x^2}{x^2 - 8x + 15} - 3 \right| = \ldots = 3 \left| \frac{8x - 15}{x^2 - 8x + 15} \right|$$

See the solution here]
